

CYD Housing v2 — Fastener Spec

ESP32-2432S028 split housing · Fusion file CYD_Housing_Split_v2 · 4 corners, all identical · 2026-09-15

BUY / PULL FROM THE BIN

Qty	Part	Spec
4	M3 × 14 mm flat-head screw 90° countersunk, hex socket	DIN 7991 / ISO 10642. Thread Ø 3.0 (measures 2.9–3.0). Head Ø 5.5–6.0, head height ≈1.65. 2 mm hex key. Length 14 is measured <i>including</i> the head. Stainless or black oxide, either is fine.
4	M3 heat-set threaded insert brass, knurled	Length 4.0 mm , outside Ø 4.0–4.2 (the “M3 × L4” size in most kits; Voron-style M3×5×4 also fits). Do not use the 5.7 mm-long ones — they bottom out in the boss.

WHICH SCREWS IN THE BIN ARE M3?

	M3	M4	M5
Thread Ø (calipers)	2.9–3.0	3.9–4.0	4.9–5.0
Flat head Ø	5.5–6.0	~8.0	~10.0
Hex key (flat head)	2.0	2.5	3.0
Drops through the CYD’s Ø3 board hole?	Yes	No	No

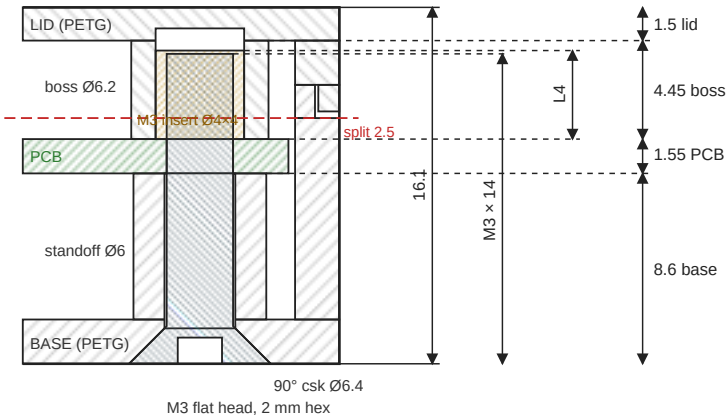
Fastest test: hold the bare board over the bin — the screw that falls through a mounting hole is M3.

HOLES IN THE PRINT (ALREADY MODELLED)

Where	Feature
Lid boss (×4)	Ø4.0 × 5.0 deep, insert pressed to flush with the boss face
Base underside (×4)	Ø3.2 through, 90° countersink Ø6.4 at the surface
Board	Ø3.0 at (±21, ±39) — screw is a clearance fit

Insert install: soldering iron with an M3 insert tip (or a clean conical tip), ~230 °C for PETG / ~200 °C for PLA. Start the insert straight, let the iron’s weight do the work, stop flush, hold 5 s, let cool before threading anything in. Do this *before* the board goes in.

SECTION THROUGH ONE CORNER (SCALE 12 PX = 1 MM)



Numbers that matter

- Thread in insert: ≈3.9 mm (tip at z 5.4, insert 1.55–5.55)
- Tip to lid face: 2.1 mm — do not substitute M3 × 16
- Torque: snug with the short arm of the key. It’s plastic.
- Seam: lid bears on the lip top; 0.3 mm parting gap at the skin

Assembly order

- Heat-set the 4 inserts into the lid bosses
- Push the light pipe up into its tube from inside
- Drop the board into the base, USB-C end first
- Lid on — lip into groove — 4 × M3 × 14 from below

Print

Base open side up · Lid face down · Light pipe flange down, clear, 100%

Section through the hole axis at one corner (all four identical). Red dash = split line z 2.5; 16.1 = overall height; L4 = insert length. Hatching: grey = printed PETG, green = PCB, gold = brass insert, blue = steel screw. Dimensions in mm.